
VORTEX

WAREHOUSE MANAGEMENT SYSTEM

Workflow Catalog & Operations Guide

VERSION 6.0 | ADVANCED INVENTORY

Sky Blue Theme | Dexie IndexedDB | React + Vite + TypeScript

PWA Enabled | Barcode/RFID Ready | Serial Number Tracking

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1. SYSTEM OVERVIEW

VortexWMS is a next-generation Warehouse Management System built as a Progressive Web App (PWA) using React, Vite, TypeScript, and Dexie (IndexedDB). It provides end-to-end inventory control from inbound receiving to outbound shipping, with advanced features including serial number tracking, directed putaway, pick-path optimization, and real-time labor analytics.

Key Capabilities

- **Multi-layer traceability** — SKU, Lot, Batch, Expiry, and Individual Serial Number tracking
- **Smart putaway** — AI-directed placement based on velocity, category, and bin constraints
- **Wave & batch picking** — Optimized pick paths with distance and time estimates
- **Real-time labor tracking** — UPH, accuracy, and distance metrics per worker
- **Integration hub** — Connect ERP, OMS, TMS, and marketplace APIs
- **PWA offline support** — Full functionality without network connectivity

Database Schema (v6)

The system uses Dexie.js with IndexedDB, currently at schema version 6. It manages 21 tables covering inventory, orders, movements, worker tasks, purchase orders, carrier rates, integration endpoints, serial numbers, and putaway tasks.

Table	Purpose	Key Indexes
inventory	Master stock records	sku, location, category, barcode, lotNumber
orders	Customer orders	orderId, status, priority, waveId, carrier
inbound	Receiving records	sku, receivedAt, poNumber, qcStatus
serialNumbers	Individual SN/IMEI tracking	serialNumber, sku, status, location
putawayTasks	Directed putaway workflow	taskId, sku, status, destinationZone
workerTasks	Assigned tasks	type, status, assignedTo, priority
cycleCounts	Audit records	sku, location, status, abcClass
purchaseOrders	PO management	poNumber, supplier, status
integrationEndpoints	External connections	name, type, status, lastSync

2. INBOUND WORKFLOW

The inbound module handles all goods entering the warehouse — from purchase order receipt through quality control to directed putaway. This is where traceability begins.

2.1 Purchase Order Receiving

When a shipment arrives, the operator scans or selects the Purchase Order (PO) number. The system displays the expected items, quantities, and any cross-dock assignments. The operator verifies physical counts against the PO, records variances, and captures lot numbers, expiry dates, and serial numbers at the point of receipt.

Workflow Steps:

1. Select PO or create ad-hoc receipt
2. Scan/enter SKU and verify quantity
3. Capture lot number, expiry, and serials (if serialized)
4. Assign receiving dock location
5. Trigger QC hold if inspection required
6. System auto-generates putaway task

TIP: Always capture lot and expiry for FEFO-controlled items (cosmetics, food, pharma). The system uses `fefoPriority` to suggest which lots to ship first.

WARN: If `expectedQty` differs from `received qty`, the system records variance and flags the PO status as "partial". Always reconcile before closing the PO.

2.2 Putaway Operations (v6)

Version 6 introduces the **PutawayTask** table — a directed workflow that guides operators from receiving dock to final storage location. The system suggests the optimal location based on zone capacity rules, item velocity, and bin constraints (weight, dimensions, temperature).

Putaway Methods:

Direct: Straight from receiving dock to final location — fastest for high-velocity items

Zone: Directed to a zone first, then final bin — used for bulk receiving

Cross-dock: Items bypass storage and go directly to outbound staging

Consolidation: Merge with existing stock at same location before putaway

TIP: Use the `zoneCapacities.putawayRules` JSON field to define custom rules. Example: `{"velocity": "high", "category": "Electronics", "maxWeight": 50}` — the system matches incoming items against these rules to suggest optimal locations.

2.3 QC Hold & Release

Items can be placed on Quality Control hold at receiving or anytime during storage. The **QCHold** table tracks SKU, lot, quantity, reason, and status. Hold statuses are: **hold** (awaiting inspection),

released (approved for stock), or **rejected** (return to supplier).

TIP: Set up automatic QC holds for new suppliers or high-value SKUs. Use the inbound.qcStatus field to flag items as pending inspection at receipt time.

3. INVENTORY MANAGEMENT

The inventory hub provides real-time visibility into stock levels, locations, and traceability. Version 6 adds full serial number lifecycle tracking for high-value items.

3.1 Stock Tracking & Traceability

Every inventory movement is logged in the `inventoryMovements` table with full context: SKU, movement type, quantity, from/to locations, operator, lot number, and timestamp. This creates an immutable audit trail for compliance and recall management.

Movement Types:

```
inbound, outbound, return, adjustment, transfer, cycle_count, qc_adjustment, putaway, pick, pack
```

TIP: Use the `logInventoryMovement()` helper to ensure every stock change is recorded. This function automatically populates the timestamp and validates the movement type against the enum.

3.2 Serial Number Management (v6)

The new `serialNumbers` table provides item-level traceability for serialized goods — essential for electronics, pharmaceuticals, and warranty-managed products. Each record tracks the full lifecycle: received → in_stock → reserved → shipped → returned.

Field	Purpose
serialNumber	Unique identifier (SN-XXXX)
imei1 / imei2	Device IMEI for mobile phones
tacPrefix	Type Allocation Code for model lookup
status	in_stock reserved shipped returned defective quarantine
lotNumber	Batch/lot association
warrantyStart / warrantyEnd	Warranty period tracking
orderId	Links to outbound order when shipped
location	Current bin location

TIP: When receiving serialized items, scan each serial number individually. The system validates against the simDb TAC prefix database to auto-populate model information.

3.3 Cycle Counting & ABC Analysis

Cycle counts are scheduled based on ABC classification. A-items (high value/velocity) are counted daily or weekly; B-items monthly; C-items quarterly. The `cycleCounts` table tracks expected vs actual quantities, variance, and approval status.

TIP: Set countFrequency and nextScheduled fields to automate cycle count scheduling. The system will highlight overdue counts on the dashboard.

4. ORDER FULFILLMENT

From order import to carrier handoff, the fulfillment module orchestrates wave creation, pick path optimization, packing, and shipping label generation.

4.1 Wave Batching & Pick Paths

Orders are grouped into waves based on priority, carrier cutoff times, and zone profiles. The pickPaths table stores optimized routes as JSON arrays: [{location, sku, qty, zone, distance}]. The system calculates totalDistance and estimatedTime for each path.

Picking Methods:

Single: One order per picker — simplest, highest accuracy

Batch: Multiple orders picked together — fastest for small items

Zone: Pickers restricted to their zone — parallel processing

Wave: Large-scale batch across all zones — highest throughput

TIP: Use batchGroups.pickingMethod to control strategy. For urgent orders, switch to "single" with high-priority flag to bypass wave batching.

4.2 Pick & Pack Operations

The workerTasks table tracks every pick, pack, and ship task. Tasks include estimatedTime, actualTime, and distance walked — enabling performance analytics. The suggestedBoxSize field on orders recommends the optimal packaging based on item dimensions and weight.

TIP: Configure boxSizes table with your actual packaging inventory. The system selects the smallest box that fits all items with padding, minimizing shipping costs.

4.3 Shipping & Carrier Selection

The carrierRates table stores rate cards per carrier, service, weight range, and zone. At ship time, the system queries active rates and recommends the lowest-cost option meeting the delivery deadline. Shipping labels are generated and stored in the shippingLabels table.

TIP: Keep carrierRates.active = true only for current contracts. Expired rates with active=false are preserved for historical reporting but excluded from selection.

5. RETURNS PROCESSING

Returns are received via the Returns module. The operator scans the original order, selects the reason (defective, wrong item, customer remorse, etc.), and decides whether to restock, quarantine, or dispose. Restocked items trigger a new putaway task.

Return Outcomes:

Restocked: Item returned to inventory — new putaway task generated

Quarantined: Item held for inspection — serial status set to quarantine

Defective: Item marked defective — serial status updated, warranty claim initiated

Disposed: Item destroyed — inventory adjusted, movement logged

TIP: For serialized returns, always scan the serial number. The system auto-updates the serialNumbers.status to "returned" and records the return timestamp for warranty tracking.

6. LABOR MANAGEMENT

The Labor Management module tracks worker performance across all task types. The workerPerformance table stores daily metrics: picksCompleted, accuracy, distanceWalked, tasksCompleted, avgTaskTime, and UPH (units per hour).

Key Metrics:

UPH (Units Per Hour): Primary productivity metric — total units picked/packed per hour

Accuracy: Percentage of tasks completed without errors or returns

Distance Walked: Meters per shift — identifies layout inefficiencies

Task Breakdown: pickCount, packCount, receiveCount, putawayCount

TIP: Use workerTasks.estimatedTime vs actualTime to identify training needs. Workers consistently exceeding estimates may need additional training on the pick path or layout.

7. INTEGRATION HUB

The Integration Hub connects VortexWMS to external systems via the `integrationEndpoints` table. Supported types: ERP, OMS, TMS, WMS, Marketplace, Carrier, and Custom. Each endpoint stores configuration as JSON, sync interval, last sync time, and error counts.

Integration Types:

ERP: SAP, Oracle, NetSuite — master data sync (SKUs, POs, vendors)

OMS: Shopify, Magento, Salesforce — order import and status updates

TMS: FedEx, UPS, DHL — rate fetching, label printing, tracking

Marketplace: Amazon, Ozon, eBay — listing sync, order routing

TIP: Monitor `integrationEndpoints.errorCount` and `lastError` fields. Set up alerts when `errorCount` exceeds a threshold or `lastSync` is older than 2x the `syncInterval`.

8. TIPS, TRICKS & BEST PRACTICES

Barcode & Scanning

- Always print barcodes with quiet zones (minimum 2.5mm on each side)
- Use Code 128 for alphanumeric SKUs — highest density and reliability
- Test scanners on glossy surfaces before bulk printing labels
- For serialized items, use QR codes to encode SN + IMEI + Lot in one scan

Zone Layout Optimization

- Place A-items (high velocity) closest to packing stations
- Group frequently co-picked items in adjacent bins to reduce travel
- Reserve floor-level bins for heavy items to prevent lifting injuries
- Use cold storage (binType: cold) for temperature-sensitive inventory

FEFO/FIFO Management

- Set fefoPriority higher for lots closer to expiry (1 = ship first, 5 = ship last)
- Run expiry reports weekly for perishable goods
- Flag items within 30 days of expiry for markdown or donation

Performance Optimization

- Keep batch group sizes under 50 orders for single-zone picking
- Assign pickers to zones where they have highest UPH
- Use wave batches for standard orders, single pick for urgent/SLA orders
- Pre-print shipping labels during wave planning to reduce pack station bottlenecks

Data Integrity

- Run exportAllData() weekly for backup — exports all 21 tables as JSON
- Use auditLogs for compliance tracking — every critical action is logged
- Validate TAC prefixes against simDb when receiving phones to catch counterfeit devices
- Reconcile purchaseOrders vs inbound weekly to catch unreceived lines

9. TROUBLESHOOTING GUIDE

Symptom	Cause	Solution
Negative stock shown	Movement logged without matching inbound	Recycle count; check inventoryMovements for unmatched
Pick path too long	Zone layout not optimized	Run ABC analysis; re-slot A-items closer to pack stations
Serial number conflict	Duplicate SN scanned	Check serialNumbers table for duplicates; validate with supplier
PO variance high	Supplier short-shipped or over-shipped	Document variance; update PO status; initiate claim if needed
Integration sync failed	Token expired or endpoint changed	Check integrationEndpoints.lastError; re-authenticate
Putaway task stuck	Suggested location full	Check zoneCapacities.currentUtilization; reassign to alternative
Worker UPH dropped	New layout or product mix	Review pickPath JSON; compare distance vs historical average
Shipping cost spike	Wrong box size selected	Review boxSizes configuration; verify item dimensions are correct
QC hold backlog	Insufficient inspection staff	Reallocate labor from low-priority putaway to QC; escalate to management
Order stuck in Pending	Wave not released or SKU out of stock	Check inventory.reorderPoint; verify waveBatch status is "open"

VortexWMS v6.0 — Advanced Inventory Management
Built with React + Vite + TypeScript + Dexie + Tailwind CSS
Sky Blue Theme | PWA Enabled | Offline Ready